FLARE: Multi-Modal Grounded Planning and Efficient Replanning for Few-Shot Embodied Agents

Anonymous Authors

Anonymous Institution

Abstract

Few-shot embodied instruction following is appealing because collecting long-horizon language-action trajectories is expensive, but current large language model (LLM) planners remain brittle in this regime: they often generate plans that are linguistically plausible yet poorly grounded in the agent's current scene. We present FLARE, a modular embodied agent that improves few-shot planning with two complementary mechanisms. First, a Multi-Modal Planner retrieves in-context demonstrations using both the task instruction and the agent's initial panoramic observation, then prompts an LLM to produce compact action-object-receptacle subgoal triplets. Second, Environment Adaptive Replanning repairs only failed subgoals during execution by replacing unseen targets with semantically similar objects detected in the current environment, without another LLM query. On ALFRED with the 0.5% few-shot protocol, FLARE with GPT-4 reaches 40.88 success rate and 51.72 goal-condition success on Test Unseen with full instructions, substantially outperforming LLM-Planner. Even without adaptive replanning, FLARE yields much stronger static planning accuracy, improving unseen planner accuracy from 30.12 to 67.48. These results indicate that multi-modal grounding and lightweight environment-aware repair are both important for robust few-shot embodied execution.

1 Introduction

Embodied agents that follow natural language instructions must jointly interpret language, perceive their surroundings, navigate in partially observed 3D scenes, and manipulate objects over long horizons. Benchmarks such as ALFRED make this challenge explicit by requiring agents to execute compositional household tasks from egocentric observations in AI2-THOR environments (Shridhar et al. 2020; Kolve et al. 2017). Strong fully supervised systems have been built with hierarchical policies, semantic maps, and explicit memory (Singh et al. 2021; Min et al. 2022; Blukis et al. 2021; Kim et al. 2023), but they typically rely on large collections of expert trajectories and instruction-action annotations.

LLM-based planners offer an alternative route for data-efficient embodied learning. By leveraging broad world knowledge and in-context learning, they can infer high-level subgoals from only a small set of examples (Ahn et al. 2022; Song et al. 2023). However, the central failure mode of few-shot LLM planning is grounding: a plan may be reasonable in the abstract while ignoring what is actually present in the current scene. In ALFRED, such errors are costly because one incorrect target object or receptacle can derail the entire trajectory.

This paper studies how to make few-shot embodied planning better grounded and cheaper to repair. Our key observation is that planning failures arise from two distinct sources. The first is *retrieval-time mismatch*: textual nearest neighbors alone may not reflect the layout or objects visible from the initial observation. The second is *execution-time mismatch*: even a sensible initial plan can include a target that is absent, occluded, or replaced by a functionally similar alternative in the current environment. These two issues call for different interventions.

FLARE addresses them with a two-stage design. The Multi-Modal Planner (MMP) retrieves demonstrations using both language similarity and similarity between initial panoramic observations, then asks the LLM to emit compact subgoal triplets of the form (action, object, receptacle). The Environment Adaptive Replanning module (EAR) runs only at execution time and only when needed: if a target remains unfound after repeated search, EAR replaces that target with the most semantically similar detected alternative without re-querying the LLM. The resulting system keeps the benefits of LLM-based high-level reasoning while reducing dependence on repeated LLM calls during online control.

Empirically, FLARE is effective in the low-data regime. In the 0.5% ALFRED setting, FLARE with GPT-4 improves Test Unseen success rate from 16.42 to 40.88 over LLM-Planner with full instructions, and from 13.41 to 32.57 in the goal-only setting. Static planner accuracy also improves consistently across all tested LLM backbones. These gains show that grounding the prompt with perception and restricting replanning to local semantic repairs are both practical design choices for few-shot embodied agents.

Our contributions are threefold. First, we introduce a multi-modal retrieval scheme for few-shot embodied planning that combines instruction and initial-scene evidence. Second, we propose an LLM-free adaptive replanning mechanism that corrects only the failing subgoal using detected environment objects. Third, we show state-of-the-art few-shot results on ALFRED using only 0.5% training data

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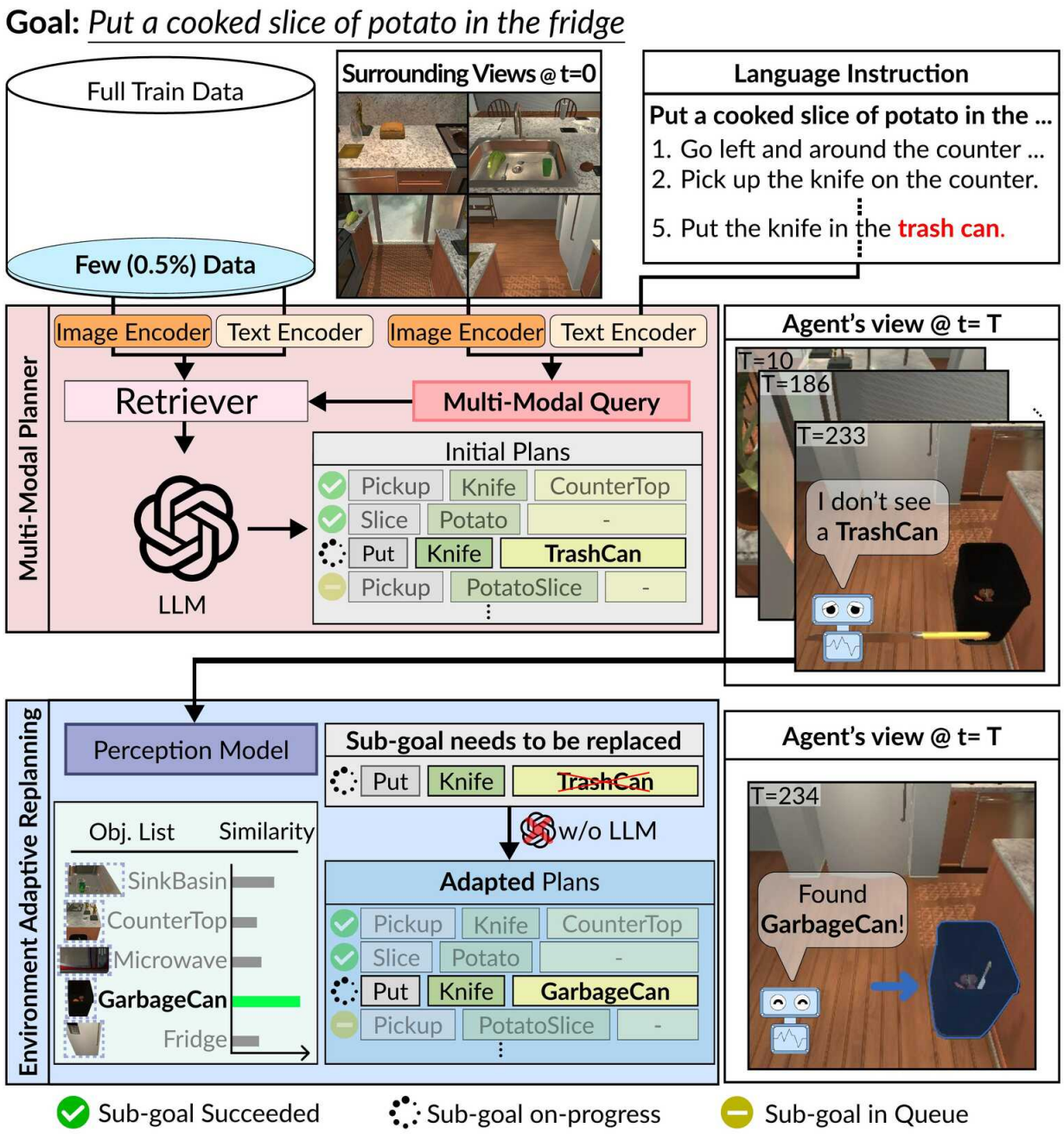

Figure 1: Overview of FLARE. MMP retrieves few-shot demonstrations using both the instruction and the initial panoramic observation to generate grounded subgoal triplets. During execution, EAR monitors the current subgoal and repairs only the failing target by replacing it with the most semantically compatible detected object, avoiding an additional LLM call.

while retaining strong interpretability through explicit subgoal triplets.

2 Related Work

2.1 Embodied Instruction Following

ALFRED has become a standard testbed for long-horizon embodied instruction following because it couples navigation, interaction, and state changes under partial observability (Shridhar et al. 2020). Early and subsequent systems improved performance through modularization, object-centric reasoning, semantic mapping, and persistent memory (Singh et al. 2021; Min et al. 2022; Blukis et al. 2021; Kim et al. 2023). These approaches demonstrate the value of structured state and explicit planning, but most are trained with full supervision. In contrast, our focus is the few-shot regime where only a small number of paired examples are available,

making the quality of high-level planning especially critical.

2.2 LLMs for Grounded Planning

Recent work has shown that large language models can act as high-level planners when combined with low-level controllers or skill libraries (Ahn et al. 2022; OpenAI et al. 2024; Touvron et al. 2023). LLM-Planner is particularly relevant because it brings few-shot LLM planning to ALFRED and demonstrates that even 0.5% training data can be useful when the planner is properly prompted (Song et al. 2023). Our method shares the motivation of exploiting LLM prior knowledge under limited data, but differs in two ways. First, we ground retrieval with scene perception rather than text alone. Second, our online correction step does not invoke the LLM again; instead, it performs local semantic replacement directly from observed object categories.

2.3 Grounding, Memory, and Adaptation

Grounding language in perceptual context has long been central in embodied AI, whether through explicit semantic maps, context-aware memory, or environment-conditioned policies (Min et al. 2022; Blukis et al. 2021; Kim et al. 2023). Test-time adaptation has also been studied in navigation and robotics, often by updating policies, uncertainty estimates, or memory over time (Wortsman et al. 2019; Ahn et al. 2022). Our approach is narrower and cheaper: EAR does not relearn a policy or regenerate the full plan. It repairs only the unsupported object or receptacle inside the active subgoal, using semantic similarity over detected candidates. This localized repair mechanism is well matched to long-horizon instruction following, where a single hallucinated target often causes failure while the rest of the plan remains valid.

3 Method

3.1 Problem Setting

We consider embodied instruction following in ALFRED (Shridhar et al. 2020), where an agent receives a task description together with optional step-by-step instructions, observes an egocentric RGB-D stream, and must execute a sequence of navigation and manipulation actions in AI2-THOR (Kolve et al. 2017). We follow the few-shot setting of LLM-Planner (Song et al. 2023), where only 0.5% of paired training trajectories are used to construct demonstrations for in-context planning.

FLARE is a modular system built on a semantic-map-based embodied agent derived from FILM and CAPEAM (Min et al. 2022; Kim et al. 2023). The LLM is used only for high-level planning. Low-level navigation is handled deterministically by a Fast Marching Method (FMM) planner over the traversible map, while perception and object interaction operate through the underlying embodied agent stack.

3.2 Multi-Modal Planner

The goal of MMP is to generate a compact, executable high-level plan that is grounded in both the instruction and the initial environment state.

Multi-modal retrieval. For each evaluation task, we form a query from two inputs: the language instruction and an initial panoramic observation created by concatenating four surrounding views at time step 0. The codebase embeds the instruction with a BERT-based sentence encoder (Devlin et al. 2018) and the panorama with CLIP visual features (Radford et al. 2021). Given a training example i, MMP computes normalized text and image similarity scores and combines them with equal weights:

$$s_i = \lambda_{\text{text}}\, \hat{s}_i^{\text{text}} + \lambda_{\text{img}}\, \hat{s}_i^{\text{img}}, \tag{1}$$

where $\lambda_{\text{text}} = \lambda_{\text{img}} = 1$ in the released implementation. We retrieve the top $k = 9$ demonstrations with the largest combined score.

This retrieval design is important because instructions alone can be ambiguous about object layout and scene affordances. Initial observations provide a weak but useful estimate of what objects, receptacles, and room configurations are likely to matter for planning.

Triplet-based plan generation. The retrieved demonstrations are inserted into an LLM prompt together with the current task. Instead of asking for free-form reasoning, MMP requests a structured sequence of subgoal triplets:

$$(A_n, O_n, R_n),$$

where A_n is a high-level action, O_n is the manipulated object, and R_n is the destination or supporting receptacle. The allowed actions in the implementation are `ToggleObject`, `CleanObject`, `HeatObject`, `PickupObject`, `SliceObject`, `CoolObject`, and `PutObject`.

This representation makes the plan explicit and easy to post-process. After generation, each triplet is expanded into executable low-level interactions using hand-coded conversions. For example, `HeatObject` becomes a sequence of opening the microwave, placing the object, toggling, and retrieving it; `PutObject` may insert open and close actions if the receptacle is openable. The resulting action list preserves a mapping from low-level actions back to high-level subgoals, which is later used by EAR.

3.3 Environment Adaptive Replanning

Initial plans can still fail when the current target is not found during execution. Regenerating the full plan with the LLM is expensive and often unnecessary because the rest of the plan may still be correct. EAR therefore performs *partial* repair.

Failure detection. During execution, the agent keeps track of the active subgoal and the set of detected objects in the current scene. If the target object or receptacle for the active subgoal is repeatedly not observed during search, the system triggers adaptive replanning. In the released code, this is implemented with a confusion counter and only activates when there is sufficient visual evidence from the current scene.

Semantic replacement without an LLM. EAR replaces only the failing argument in the current subgoal. Suppose the current target category is c and the detected object set is

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$\mathcal{O}$. We encode c and the candidate detected categories with a BERT sentence encoder (Devlin et al. 2018), then choose

$$\tilde{c} = \arg\max_{o \in \mathcal{O}} \cos\big(f(c), f(o)\big), \tag{2}$$

where $f(\cdot)$ denotes the text embedding. Separate embedding banks are used for pickupable objects and receptacles, matching the action type of the current subgoal. The updated category replaces the original argument in the active plan, and execution resumes immediately. Importantly, EAR does not alter previously completed subgoals and does not regenerate unaffected future steps.

Figure 2 shows two typical outcomes. In one case, MMP produces a better initial cleaning plan than a text-only variant. In another, EAR recovers from an invalid receptacle target by switching to a visually observed alternative.

3.4 Execution Policy

Once the subgoal list is available, the embodied agent executes it with a deterministic low-level controller. The environment state is summarized in a semantic map, object detections, and depth observations as in prior modular ALFRED agents (Min et al. 2022; Kim et al. 2023). Navigation between interaction points uses an FMM planner on the traversible map. This separation lets FLARE concentrate LLM usage on high-level planning while relying on robust classical planning for short-horizon motion.

4 Experiments

4.1 Experimental Setup

Dataset and protocol. We evaluate on ALFRED (Shridhar et al. 2020) in AI2-THOR (Kolve et al. 2017). Following the few-shot protocol used by LLM-Planner (Song et al. 2023), we use 0.5% training data to construct in-context examples and report results on Test Seen and Test Unseen. We evaluate two language settings: *goal instruction + sequential instructions* and *goal instruction only*.

Metrics. We report Success Rate (SR) and Goal-Condition Success (GC), together with their path-length-weighted (PLW) versions when available. We also measure planner accuracy on valid seen and valid unseen folds by checking whether the predicted plan matches the ground-truth plan. For this static planning evaluation, both LLM-Planner and FLARE are tested without online replanning.

Backbones and baselines. We instantiate FLARE with four LLM backbones: LLaMA2, Vicuna, GPT-3.5, and GPT-4. We compare against few-shot baselines HLSM, FILM, CAPEAM, and LLM-Planner, as well as full-data HLSM, FILM, and CAPEAM where reported in prior work (Blukis et al. 2021; Min et al. 2022; Kim et al. 2023; Song et al. 2023).

4.2 Main Results

Table 1 summarizes the main embodied execution results. FLARE consistently improves over LLM-Planner across all LLM backbones, with the largest gains appearing on the more challenging unseen split. With GPT-4 and full instructions, FLARE reaches 40.88 SR and 51.72 GC on Test Unseen, compared with 16.42 SR and 23.37 GC for LLM-Planner. The improvement is similarly large in the goal-only setting, where FLARE-GPT-4 obtains 32.57 SR and 43.23 GC on Test Unseen.

These gains are notable for two reasons. First, they are achieved in the same 0.5% regime where prior non-LLM few-shot baselines perform poorly. Second, the PLW metrics remain strong, indicating that the improvement is not coming from overly long or inefficient trajectories alone. Among the tested backbones, stronger LLMs produce better plans overall, but FLARE improves performance even for weaker open models, suggesting that better grounding complements model scale rather than merely replacing it.

4.3 Static Planner Accuracy

Table 2 isolates high-level planning quality by removing online replanning. Here the effect of MMP is clear. FLARE without EAR dramatically outperforms the static version of LLM-Planner for every tested backbone. With GPT-4, unseen planner accuracy increases from 30.12 to 67.48, and seen accuracy from 31.54 to 61.34. The same pattern holds for GPT-3.5, Vicuna, and LLaMA2.

This result supports the main claim behind MMP: retrieving demonstrations with both language and initial visual context leads to plans that are better aligned with the actual task and scene. The gap is especially large for weaker backbones, which are more sensitive to prompt quality.

5 Analysis

5.1 Ablation Study

Table 3 evaluates the contribution of each component using GPT-3.5. Removing either MMP or EAR lowers performance relative to the full model, and removing both causes the largest degradation. On Test Unseen SR, the full system reaches 31.79, while removing MMP yields 30.35, removing EAR yields 30.28, and removing both yields 28.58.

The ablation indicates that the two modules are complementary. MMP improves the quality of the initial plan before any action is taken, while EAR helps when the initial plan is locally incompatible with the current environment. The fact that both single ablations remain above the configuration without either module suggests that each component contributes independently rather than only through one shared mechanism. This complementarity is also visible in unseen GC: starting from 39.92 without either module, EAR alone reaches 42.40, MMP alone reaches 42.48, and enabling both raises GC further to 43.94.

5.2 Qualitative Analysis

Figure 2(a) illustrates the benefit of multi-modal retrieval. For the instruction "Put clean soap on the counter," a weaker planner without MMP generates an incorrect plan that attempts to place the soap in the sink basin rather than clean it and then move it to the countertop. FLARE instead produces the correct sequence of pickup, clean, and place subgoals. This example is consistent with the planner-accuracy gains in Table 2: better exemplars lead to better decomposition.

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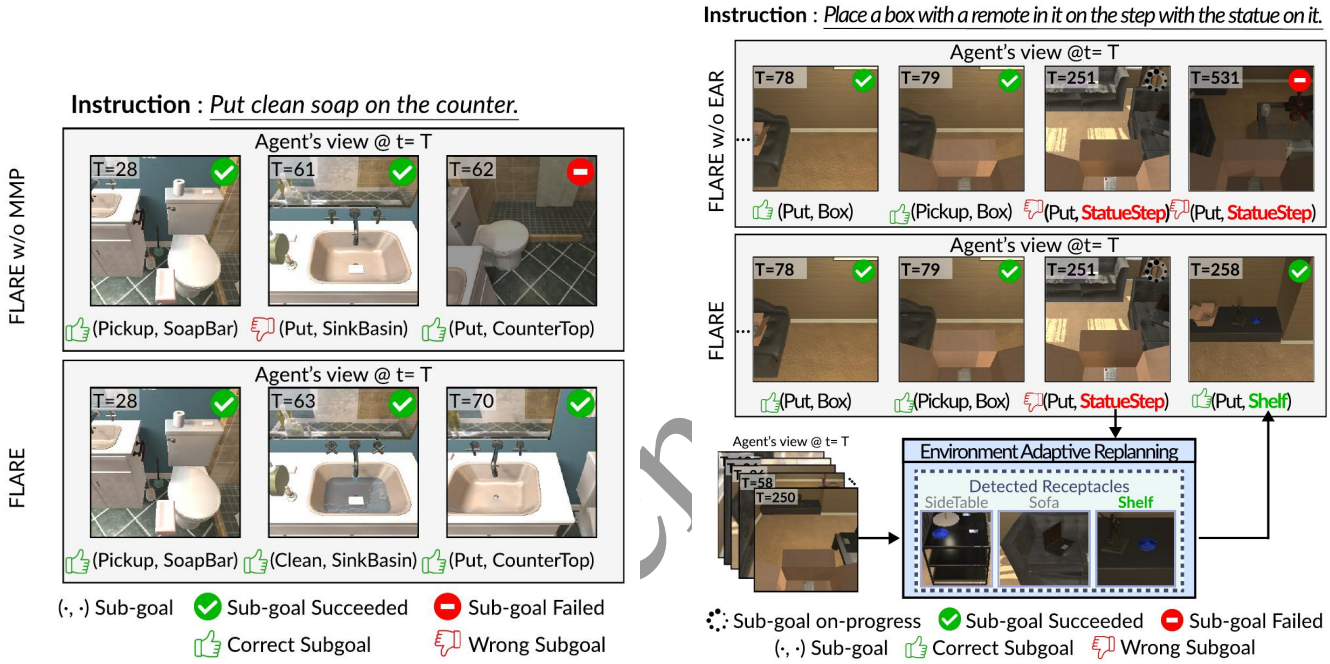
Table 1: **Comparison with state-of-the-art methods.** The path-length-weighteed (PLW) metrics are presented in the parentheses for each metric. †We excerpt 'SR' and 'GC' from (Song et al. 2023). For models that did not report the PLW metric, we noted 'N/A' in our comparison.

Setting	Model	Goal instructions + Sequential instructions				Goal instruction only			
		Test Seen		Test Unseen		Test Seen		Test Unseen	
		SR	GC	SR	GC	SR	GC	SR	GC
Few-shot (0.5%)	HLSM (Blukis et al. 2021)†	0.82 (N/A)	6.88 (N/A)	0.61 (N/A)	3.72 (N/A)	N/A	N/A	N/A	N/A
	FILM (Min et al. 2022)†	0.00 (N/A)	4.23 (N/A)	0.20 (N/A)	6.71 (N/A)	N/A	N/A	N/A	N/A
	CAPEAM (Kim et al. 2023)	0.00 (0.00)	3.90 (2.29)	0.20 (0.00)	6.63 (2.36)	N/A	N/A	N/A	N/A
	LLM-Planner (Song et al. 2023)	18.20 (N/A)	26.77 (N/A)	16.42 (N/A)	23.37 (N/A)	15.33 (N/A)	24.57 (N/A)	13.41 (N/A)	22.89 (N/A)
	FLARE-LLaMA2	16.96 (4.60)	24.84 (8.09)	17.79 (5.62)	27.40 (9.46)	12.00 (3.01)	20.05 (7.22)	13.73 (4.27)	21.98 (8.46)
	FLARE-Vicuna	20.61 (6.28)	29.57 (10.17)	22.04 (7.61)	33.57 (12.06)	16.37 (4.57)	23.68 (8.84)	18.05 (5.98)	26.75 (10.75)
	FLARE-GPT-3.5	32.55 (12.17)	42.02 (16.94)	31.79 (12.21)	43.94 (17.44)	23.48 (8.71)	33.40 (14.40)	25.38 (9.37)	36.02 (15.28)
	FLARE-GPT-4 (Ours)	**40.05 (16.68)**	**48.84 (21.31)**	**40.88 (18.14)**	**51.72 (22.78)**	**31.96 (12.93)**	**41.36 (18.55)**	**32.57 (12.72)**	**43.23 (18.40)**
Full	HLSM (Blukis et al. 2021)	29.94 (8.74)	41.21 (14.58)	20.27 (5.55)	30.31 (9.99)	25.11 (6.69)	35.79 (11.53)	16.29 (4.34)	27.24 (8.45)
	FILM (Min et al. 2022)	28.83 (11.27)	39.55 (15.59)	27.80 (11.32)	38.52 (15.13)	25.77 (10.39)	36.15 (14.17)	24.46 (9.67)	34.75 (13.13)
	CAPEAM (Kim et al. 2023)	51.79 (21.60)	60.50 (25.88)	46.11 (19.45)	57.33(24.06)	47.36 (19.03)	54.38 (23.78)	43.69 (17.64)	55.66 (22.76)



(a) MMP corrects the high-level subgoal sequence for a cleaning task, producing `CleanObject` rather than an incorrect direct placement plan.

(b) EAR repairs an invalid receptacle target online by replacing `StatueStep` with the visually observed `Shelf`.

Figure 2: Qualitative examples of the two components of FLARE. MMP yields a better initial plan from the same instruction, while EAR repairs only the failing subgoal argument using objects detected in the current environment.

Figure 2(b) highlights why local online repair matters. The initial plan contains a receptacle target (`StatueStep`) that is not supported by the observed scene. Rather than regenerating the full plan, EAR scans the currently detected receptacles and substitutes the semantically closest viable alternative, `Shelf`. The repaired plan succeeds because only the target argument was wrong; the rest of the trajectory remains useful.

Taken together, the quantitative and qualitative results suggest a division of labor. MMP improves global task decomposition, while EAR improves local robustness to environment mismatch. This separation is particularly attractive for few-shot embodied control because it avoids repeated expensive LLM interaction during online execution.

Generalization to unseen scenes. A consistent pattern is that the gains from better grounding are strongest on harder generalization settings. In static planning with GPT-4, unseen accuracy improves by 37.36 points, from 30.12 to 67.48, while seen accuracy improves by 29.80 points, from 31.54 to 61.34. The same trend appears in embodied execution: with full instructions, GPT-4 improves Test Unseen SR by 24.46 points over LLM-Planner, compared with 21.85 points on Test Seen. This suggests that the visual retrieval signal is especially useful when language alone underspeci-

Table 2: **Planner accuracy comparison.** 'Seen Acc.' and 'Unseen Acc.' denote planner accuracies in valid seen and unseen folds. A plan is regarded as correct when it is aligned with the ground-truth plan. To solely compare planner accuracy, we omit replanning in LLM-Planner and FLARE, denoted with 'Static' and 'w/o EAR'.

LLM	Method	Seen Acc.	Unseen Acc.
LLaMA2	LLM-Planner (Static) (Song et al. 2023)	0.006	0.002
LLaMA2	**FLARE (w/o EAR)**	**18.54**	**22.29**
Vicuna	LLM-Planner (Static) (Song et al. 2023)	8.17	7.06
Vicuna	**FLARE (w/o EAR)**	**24.51**	**33.62**
GPT-3.5	LLM-Planner (Static) (Song et al. 2023)	29.78	31.67
GPT-3.5	**FLARE (w/o EAR)**	**46.10**	**55.66**
GPT-4	LLM-Planner (Static) (Song et al. 2023)	31.54	30.12
GPT-4	**FLARE (w/o EAR)**	**61.34**	**67.48**

fies scene layout or object availability.

Behavior across LLM backbones. The benefit of better grounding is visible across all tested planners rather than only the strongest model. On Test Unseen with full instructions, SR rises from 16.42 for LLM-Planner to 17.79 with FLARE-LLaMA2, 22.04 with FLARE-Vicuna, 31.79 with FLARE-GPT-3.5, and 40.88 with FLARE-GPT-4. Static planning shows the same ordering: unseen accuracy reaches 22.29, 33.62, 55.66, and 67.48 for the four backbones, respectively. This pattern suggests that MMP provides a cleaner task-scene interface that scales with model capability, while still reducing prompt mismatch for weaker open models.

Efficiency implications. Together with EAR, this scaling trend is practically useful: stronger models exploit better retrieval, while weaker ones still avoid unnecessary full-plan regeneration when failures are only local.

From plans to execution. The gap between planner accuracy and embodied SR indicates that better plans are necessary but not sufficient for successful long-horizon control. EAR is important because it converts part of MMP's planning gains into realized task completion when test-time failures come from local scene mismatch rather than global task misunderstanding.

6 Conclusion

We presented FLARE, a few-shot embodied agent that combines multi-modal in-context retrieval with lightweight environment-adaptive replanning. The method improves grounding at two stages: before execution, by retrieving demonstrations using both instructions and initial panoramic observations; and during execution, by repairing only unsupported subgoals through semantic replacement over detected objects. On ALFRED, this design yields large gains over prior few-shot planning baselines and substantially improves static planner accuracy across multiple LLM backbones. The results show that robust few-shot embodied planning benefits not only from stronger language models, but also from explicitly grounding prompts in perception and constraining replanning to the smallest necessary correction. Because EAR edits only the active subgoal argument, it also avoids repeated full-plan regeneration during rollout, preserving interpretability while keeping online adaptation lightweight. This makes the approach especially practical when online latency or LLM query budgets are constrained, and the modular design keeps retrieval, planning, and repair easy to diagnose.

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Table 3: **Ablation study.** PLW metrics are presented in the parentheses for each metric. MMP and EAR each denotes 'Multi-Modal Planner' and 'Environment Adaptive Replanning,' respectively. We observe that each component contributes to agent's performance.

#	MMP	EAR	Test Seen		Test Unseen	
			SR	GC	SR	GC
(a)	✓	✓	32.55 (12.17)	42.02 (16.94)	31.79 (12.21)	43.94 (17.44)
(b)	✗	✓	30.20 (12.13)	41.26 (17.27)	30.35 (11.62)	42.40 (16.66)
(c)	✓	✗	30.79 (11.98)	40.20 (16.51)	30.28 (12.01)	42.48 (17.03)
(d)	✗	✗	28.05 (11.48)	38.64 (16.23)	28.58 (11.82)	39.92 (16.13)

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